

MARA LEVY

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EDUCATION

University of Maryland

Ph.D. Candidate expected graduation May 2025

Advisor: Abhinav Shrivastava

College Park, MD

September 2019 - Current

University of Maryland

M.S. Computer Science

Advisor: Abhinav Shrivastava

College Park, MD

September 2019 - May 2021

University of Pennsylvania

School of Engineering and Applied Science - Bachelor of Science in Engineering

Major: Computer and Information Science

Minor: Mathematics

Cumulative GPA: 3.84 / 4.00 (Summa Cum Laude and Tau Beta Pi)

Philadelphia, PA

May 2019

PUBLICATIONS

P3-PO: Prescriptive Point Priors for Visuo-Spatial Generalization of Robot Policies

Mara Levy, Siddhant Haldar, Lerrel Pinto, Abhinav Shrivastava

ICRA 2025

TREND: Tri-teaching for Robust Preference-based Reinforcement Learning with Demonstrations

Shuaiyi Huang, Mara Levy, Anubhav Gupta, Daniel Ekpo, Ruijie Zheng, Abhinav Shrivastava

ICRA 2025

VeriGraph: Scene Graphs for Execution Verifiable Robot Planning

Daniel Ekpo, Mara Levy, Saksham Suri, Chuong Huynh, Abhinav Shrivastava

Under Submission

NeRF-Aug: Data Augmentation for Robotics

with Neural Radiance Fields

Eric Zhu, Mara Levy, Matthew Gwilliam, Abhinav Shrivastava

Under Submission

Active Region Video Diffusion for Universal Policies

Shuaiyi Huang, Mara Levy, Zhenyu Jiang, Anima Anandkumar, Yuke Zhu, Linxi Fan, De-An Huang, Abhinav Shrivastava

IROS 2024

WayEx: Waypoint Exploration using a Single Demonstration

Mara Levy, Nirat Saini, Abhinav Shrivastava

ICRA 2024

Coarse-to-Fine Human Mesh Recovery with Transformers

Vatsal Agarwal, Mara Levy, Max Ehrlich, Youbao Tang, Ning Zhang, Ruei-Sung Lin, Abhinav Shrivastava

ECCV Workshop 2025

V-UIPE: Variational View Invariant Pose Embedding

Mara Levy, Abhinav Shrivastava

CVPR Workshop 2024

No-frills Dynamic Planning using Static Planners

Mara Levy, Vasista Ayyagari, Abhinav Shrivastava

ICRA 2021

RESEARCH EXPERIENCE

University of Maryland Institute for Advanced Computer Studies

Graduate Research Assistant

College Park, MD (2019 -)

Advisor: Abhinav Shrivastava

- Addressed robotic interaction with dynamic environments where environment changes are independent of the agent
- Worked on a reinforcement learning based approach to learning new skills from a single demonstration
- Utilized waypoint information to perform imitation learning from a small number of examples on a wide range of tasks

New York University Computational Intelligence, Learning, Vision and Robotics Lab

Visiting Research Assistant

New York, NY (2024 - 2024)

Advisor: Lerrel Pinto

- Worked on a wide range of tasks concerning the physical robot to gain more experience with controls.
- Assisted the work of several other students as well as conducting my own research project.

University of Pennsylvania: GRASP Lab*Undergraduate Research Assistant*

- Used temporal video data to improve current key point detection done by "6-DoF Object Pose from Semantic Key-points"
- Utilized improved data to retrain the model to produce better pose estimates

Philadelphia, PA (2018 - 2019)*Advisor: Kostas Daniilidis***University of Pennsylvania: "Crowdsourcing and Human Computation"***Undergraduate Research Assistant*

- Used crowd-sourced data and programmatic analysis to identify and summarize key attributes of privacy policies
- Won best final project of 40+ projects and received \$10,000 for further research
- Paper was accepted as a work in progress paper to HCOMP 2017

Philadelphia, PA (2017 - 2018)*Advisor: Chris Callison-Burch*

PROFESSIONAL EXPERIENCE

Amazon*Research Scientist Intern***Arlington, VA***May 2022 - November 2022*

- Worked as a part of the Alexa Prize Team
- Researched Human Pose Estimation in order to enable Alexa devices to correct pose when teaching a skill

Flatiron Health*Software Engineering Intern***New York City, NY***June 2018 - August 2018*

- Worked as a part of the Learning Healthcare Platform team to improve data flow within the company
- Wrote an API to allow engineers to document metrics from their pipelines into InfluxDB
- Designed a Flask app to query information about the current state of pipelines and help engineers debug

Google*Software Engineering Intern***Mountain View, CA***May 2017 - August 2017*

- Worked as a part of the Google Lens Team on Google's Android Search App (AGSA)
- Wrote code to efficiently download images into AGSA and analyzed those images within the app
- Assisted in writing a server that connects the mobile application to a shared backend

HONORS AND ACHIEVEMENTS

- University of Maryland Dean's Fellowship
- Tau Beta Pi

RELEVANT COURSEWORK

- **Graduate:** Robotic Decision Making, Advanced Techniques in Visual Learning and Recognition, AI Planning, Computer Graphics
- **Undergraduate:** Data Structures & Algorithms, Linear Algebra, Computer Organization & Hardware, Crowdsourcing & Human Computation, Probability, Machine Perception